

Lecture Notes: Hypothesis Testing and Analysis of Variance

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1 Hypothesis Testing

Suppose we are given two independent samples

$$S_X = \{X_1, \dots, X_m\}, \quad S_Y = \{Y_1, \dots, Y_n\},$$

where

$$X_i \sim f_X, \quad Y_i \sim f_Y.$$

Our goal is to determine whether the two populations differ with respect to some characteristic of interest. To do so, we define a *test statistic*

$$T = T(S_X, S_Y),$$

which measures the discrepancy between the two samples.

For example, if we want to compare the means of the two populations, we might choose

$$T(S_X, S_Y) = |\hat{\mu}_X - \hat{\mu}_Y|,$$

where $\hat{\mu}_X$ and $\hat{\mu}_Y$ denote the sample means.

We formulate two competing hypotheses:

H_0 : The populations do not differ with respect to the quantity of interest,

H_1 : The populations do differ.

For example, when comparing means,

$$H_0 : \mu_X = \mu_Y,$$

$$H_1 : \mu_X \neq \mu_Y.$$

After collecting data, we compute the observed value of the test statistic

$$T(S_X, S_Y) = c.$$

A natural quantity of interest would be

$$P(H_0 \mid T = c),$$

that is, the probability that the null hypothesis is true given the observed data.

Unfortunately, classical hypothesis testing does not directly compute this quantity. Using Bayes' theorem,

$$P(H_0 \mid T) = \frac{P(T \mid H_0)P(H_0)}{P(T)},$$

and both $P(H_0)$ and $P(T)$ are generally unknown.

Instead, classical statistics evaluates how surprising the observed statistic would be if the null hypothesis were true.

2 p-Values

The key quantity is the *p-value*.

For a one-sided test,

$$p = P(T \geq c \mid H_0).$$

For a two-sided test,

$$p = P(|T| \geq |c| \mid H_0).$$

The p-value is therefore the probability of observing a test statistic at least as extreme as the one obtained from the data, assuming that the null hypothesis is true.

A small p-value indicates that the observed result would be unlikely under the null hypothesis.

Given a significance level α (commonly $\alpha = 0.05$), we reject the null hypothesis whenever

$$p \leq \alpha.$$

3 The Null Distribution

To compute a p-value, we need the distribution of the test statistic under the null hypothesis.

This distribution is called the *null distribution*:

$$F_{T|H_0}.$$

In classical statistics, null distributions are derived analytically under certain assumptions. Common examples include:

- Student's *t*-test,
- Wald test,
- χ^2 -test,
- *F*-test.

However, null distributions can also be approximated numerically using permutation methods.

4 Permutation Tests

Permutation tests provide a flexible, non-parametric approach to hypothesis testing.

Suppose we combine both samples into a pooled sample

$$S_Z = \{Z_1, \dots, Z_{m+n}\} = \{X_1, \dots, X_m, Y_1, \dots, Y_n\}.$$

Under the null hypothesis, the labels “belongs to group *X*” and “belongs to group *Y*” should not matter. Therefore, we may randomly shuffle these labels.

Algorithm

1. Compute the observed statistic

$$T_{\text{obs}} = \frac{1}{m} \sum_{i=1}^m Z_i - \frac{1}{n} \sum_{i=m+1}^{m+n} Z_i = c.$$

2. Randomly permute the pooled observations.
3. Split the permuted sample into two groups of sizes m and n .
4. Compute the statistic for the permuted data.
5. Repeat the procedure many times, e.g.,

$$k = 10\,000.$$

6. Estimate the p-value by

$$\hat{p} = \frac{1}{k} \sum_{i=1}^k \mathbf{1}(T_{\text{perm}}^{(i)} \geq c).$$

The empirical distribution of the permuted statistics approximates the null distribution.

Remarks

- If the statistic is based on a difference of means, the corresponding classical parametric test is Student's t -test.
- Under the null hypothesis, p-values are uniformly distributed on the interval $[0, 1]$.

5 Type I and Type II Errors

Every hypothesis test can lead to two types of mistakes.

	Accept H_0	Reject H_0
H_0 true	Correct	Type I error
H_0 false	Type II error	Correct

Type I Error

A Type I error occurs when we reject a true null hypothesis.

Its probability equals

$$P(\text{Type I error}) = \alpha.$$

Type II Error

A Type II error occurs when we fail to reject a false null hypothesis.

Its probability is denoted by

$$\beta.$$

Statistical Power

The *power* of a test is defined as

$$\text{Power} = 1 - \beta.$$

Power measures the probability of correctly detecting a genuine effect.

A test has higher power when

- the effect size is larger,
- the sample size is larger,
- the noise level is smaller.

Effect Size

For comparing means, a commonly used effect size is *Cohen's d*:

$$d = \frac{\mu_1 - \mu_2}{\sigma},$$

where σ denotes the common standard deviation.